

Cohort Retention Analysis

SQL Project — Online Retail Dataset

What is this project?

This project answers one simple business question:

"After a customer buys from us for the first time — do they come back?"

Using only SQL, I analyzed real transaction data from an online retail store (Dec 2010 to Dec 2011). I grouped customers by the month they first purchased (called a cohort), then tracked how many of them returned each month after that.

The final result is a retention table — a number that tells us: out of 100 customers who joined in January, how many came back in February, March, April... and so on.

The Dataset

The data used is the UCI Online Retail Dataset — a real-world e-commerce dataset from a UK-based store.

Dataset	UCI Online Retail Dataset
Time period	December 2010 — December 2011
Key columns	CustomerID, InvoiceDate, InvoiceNo, Quantity, UnitPrice
Total cohorts	13 monthly cohorts analyzed
SQL tool	PostgreSQL

How It Works — In Simple Words

Here is how I built the analysis step by step, explained without jargon:

Step 1 — Find when each customer first bought

I used `MIN(InvoiceDate)` to find the earliest purchase date for every customer. This tells us: "When did this person start?"

Step 2 — Group customers by their joining month

I used `DATE_TRUNC('month', ...)` to convert exact dates into just the month. So customers who first bought on Jan 3, Jan 15, Jan 27 all belong to the "January cohort."

Step 3 — Attach the joining month to every transaction

Using a window function (`MIN OVER PARTITION BY`), I added each customer's joining month to all their future transactions — not just the first one. Now every row in the data knows both when the customer joined AND when they purchased.

Step 4 — Calculate how many months later they returned

I used `AGE()` and `EXTRACT(MONTH)` to calculate the gap between the joining month and the purchase month. So if someone joined in January and bought again in March, that gap is 2 months.

Step 5 — Count returning users and calculate percentage

I grouped the data by cohort month and month gap, then counted unique customers. Finally, I divided by the original cohort size to get the retention percentage.

Results — Cohort Summary

Cohort	Users Joined	Month 1 Return	Month 3 Return	Best Month	Key Insight
Dec 2010	948 users	38.2%	38.7%	50% (M11)	Strongest cohort — loyalty grows over time
Jan 2011	421 users	24.0%	24.2%	36.8% (M10)	Steady mid-range performers
Feb 2011	380 users	24.7%	27.9%	31.3% (M9)	Decent retention, dropped at end
Mar 2011	440 users	19.1%	21.8%	28.9% (M8)	Started slow, recovered slightly
Jun 2011	235 users	20.9%	27.2%	33.6% (M5)	Small cohort, surprisingly strong at M5
Aug 2011	167 users	25.1%	25.1%	25.1% (M1-3)	Smallest cohort, consistent but low
Sep 2011	298 users	29.9%	12.1%	32.6% (M2)	Recovered from mid-year slump
Nov 2011	321 users	13.4%	N/A	13.4% (M1)	Too recent — limited data

Key Insights

Insight 1 — December 2010 is the power cohort

This cohort had 948 users — the largest of all 13 cohorts. What makes it truly special is that its retention actually increases over time, reaching 50% by month 11. This means half the customers who joined in December 2010 were still buying almost a year later. These are the store's most loyal, long-term customers.

Insight 2 — Month 1 churn is the biggest problem

Across every single cohort, the biggest drop always happens in month 1. Customers go from 100% to roughly 20–38% in just one month. This means 60–80% of new customers never come back after their first purchase. From a business perspective, this is where the most value can be recovered — through better onboarding, welcome emails, or follow-up discounts.

Insight 3 — Mid-2011 was the weakest period

The June, July, and August 2011 cohorts were not only the smallest (167–235 users) but also had lower long-term retention. This could indicate a seasonal slowdown, lower marketing quality, or that the customers acquired during this period were less engaged with the brand.

Insight 4 — September 2011 shows recovery

After the mid-year slump, the September cohort bounced back to 298 users and showed improved month-1 retention of 29.9%. This suggests either a seasonal demand pickup (holiday season approaching) or a better customer acquisition strategy resuming.

Insight 5 — Loyal customers exist in every cohort

Users who survive past month 3 tend to stabilize around 20–30% retention for the rest of the year. This means there is a core loyal base in every cohort. The business should identify what these customers have in common and use that to improve early-stage retention.

Business Value of This Analysis

Cohort retention analysis is one of the most important metrics for any subscription or e-commerce business. Here is why this project matters:

- Retention rate directly predicts long-term revenue and customer lifetime value (LTV)
- Month-1 drop rate shows where the business is losing the most customers — fixing this has the highest ROI
- Cohort comparison reveals whether newer customers behave differently from older ones
- Seasonal patterns in cohort size help plan marketing and inventory budgets
- This type of analysis is regularly used by analysts at companies like Amazon, Flipkart, and Swiggy to make product and marketing decision

SQL Skills Demonstrated

Window Functions	MIN() OVER (PARTITION BY) to attach cohort month to all rows
CTEs	Two chained CTEs to organize cohort data and counts cleanly
Self Join	Joining cohort table to itself to get month-0 size for % calculation
Date Functions	DATE_TRUNC, AGE, EXTRACT for date grouping and difference
Aggregation	COUNT(DISTINCT CustomerID) and GROUP BY for cohort sizing
Business Logic	Retention % = (users at month N / users at month 0) × 100